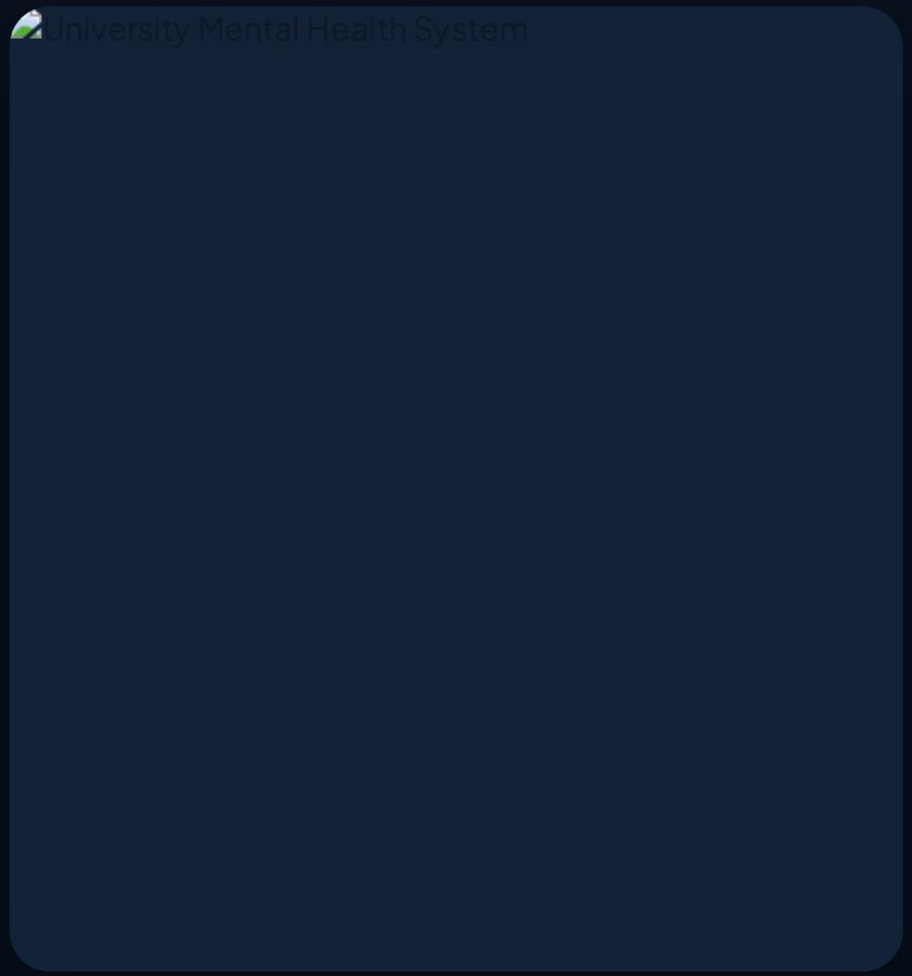
SYSTEM ARCHITECTURE & STRATEGIC FRAMEWORK

Digital Mental Health & Psychological Support System

Engineering an accessible, stepped-care digital ecosystem for higher education student wellbeing.



BASELINE REALITY

The Higher Education Care Gap

1 in 3

Students report clinical psychological distress

60%+

Never seek physical on-campus counseling

24/7

Unmet demand for off-hours crisis support

DESIGN INTENT

System Design Objectives

Stigma Reduction

Confidential barrier-free digital entry.

Immediate Triage

Real-time urgency classification.

Clinician Offloading

Automated mild symptom self-care.

CLINICAL MODEL

Stepped Care Support Hierarchy

Level 1

Psychoeducation

Self-guided wellness modules.

Level 2

Digital iCBT

Structured cognitive exercises.

Level 3

Peer Coaching

Moderated student forums.

Level 4

Clinical Telehealth

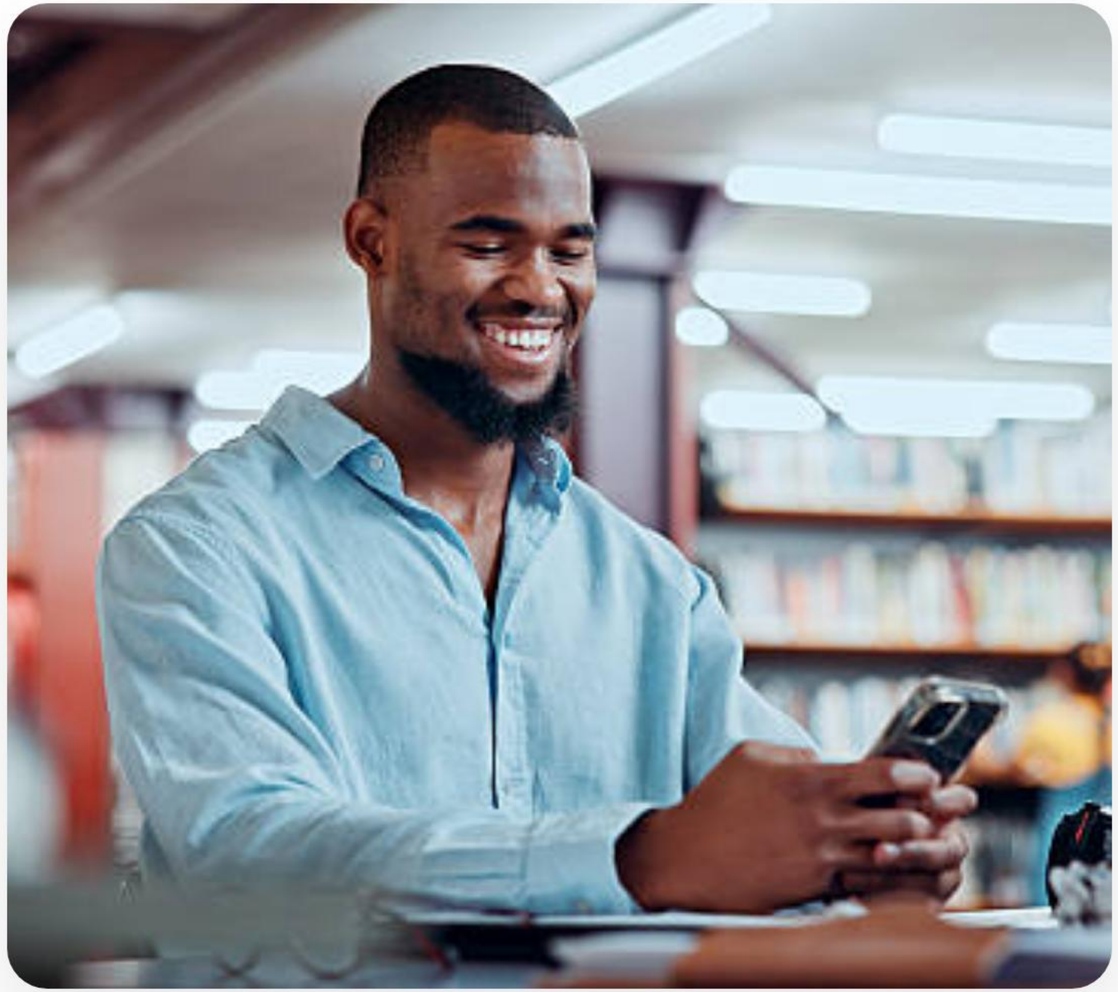
Direct licensed counseling.

USER ENGAGEMENT

Frictionless Student Experience

Campus-authenticated mobile access empowers students to log moods, access crisis hotlines, or book sessions in under three taps.

- **Anonymous daily check-ins**
- **Single Sign-On (SSO) campus sync**



SYSTEM STACK

Frontend & Clinical Engine

Student Client Interface

Multiplatform: Native iOS, Android, and web.

Micro-Interventions: Five-minute interactive modules.

Emergency Redial: One-touch crisis hotline routing.

Clinical Provider Portal

Risk Flagging: NLP-driven high-risk symptom detection.

EMR Integration: Direct campus clinic record sync.

Cohort Analytics: Anonymized stress trend heatmaps.

DATA GOVERNANCE

Privacy, Ethics & Compliance

FERPA & HIPAA

Strict statutory health compliance.

End-to-End Encryption

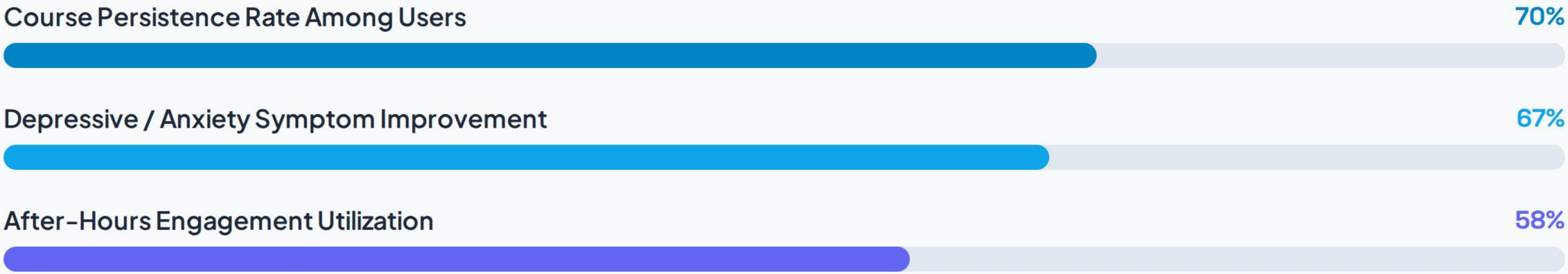
AES-256 standard data protection.

Pseudonymization

Academic records kept isolated.

PROGRAM OUTCOMES

Student Retention & Symptom Reduction



DEPLOYMENT STRATEGY

Campus Rollout Roadmap

Q1

Campus Pilot

Testing across 1,000 students.

Q2

Clinic Integration

Counseling center sync.

Q3

Campus Launch

Full student body orientation.

Q4

Impact Review

Efficacy data evaluation.

ACADEMIC INQUIRY

Thank You & Discussion

Advancing digital student wellbeing, early intervention, and institutional health infrastructure.

wellness.research@university.edu

Department of Psychological Sciences

Image Sources



https://smudailycampus.com/wp-content/uploads/2026/10/unknown-800×1200.png

Source: smudailycampus.com



https://media.istockphoto.com/id/2229544528/photo/smile-black-man-and-student-with-phone-in-library-for-social-media-reading-chat-and.jpg?s=612×612&w=0&k=20&c=RuWwefns3lXer_MSU2o2n6mlj0GXQLqUsVFA8Wl5G7Y=

Source: www.istockphoto.com